

LECTURE SHEET

EXERCISE-I

(Magnetic field due to current carrying conductors)

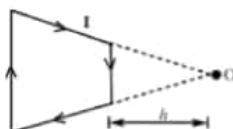
LEVEL-I (MAIN)

Straight Objective Type Questions

- An electric current passes through a long straight wire. At a distance 5 cm from the wire, the magnetic field is B. The field at 20 cm from the wire would be
 - 2B
 - $\frac{B}{4}$
 - $\frac{B}{2}$
 - B
- A current of $1/4\pi$ amp is flowing in a long straight conductor. The line integral of magnetic induction around a closed path enclosing the current carrying conductor is
 - 10^{-7} webre/metre
 - $4\pi \times 10^{-7}$ webre/metre
 - $16\pi^2 \times 10^{-7}$ webre/metre
 - zero
- A wire in the form of a square of side a carries a current i . Then the magnetic induction at the centre of the square is (magnetic permeability of free space $= \mu_0$)
 - $\frac{\mu_0 i}{2\pi a}$
 - $\frac{\mu_0 i \sqrt{2}}{\pi a}$
 - $\frac{2\sqrt{2}\mu_0 i}{\pi a}$
 - $\frac{\mu_0 i}{\sqrt{2}\pi a}$
- Due to a straight current carrying conductor, a null point occurred at p on east of the conductor. The net magnetic induction at a point Q which is half the distance of 'p' on north of the conductor is,
 - zero
 - B_H
 - $\sqrt{2} B_H$
 - $\sqrt{5} B_H$

Numerical Value Type Questions

- A current of $I = \sqrt{2}$ A flows in a circuit having the shape of an isosceles trapezium. The ratio of the bases of the trapezium is 2. The length of the smaller base of the trapezium is $l = 100$ mm, the distance $h = 50$ mm. The magnetic induction B at symmetric point O in the plane of the trapezium is $x\mu T$. Then 'x' is



- Two long parallel wires are separated by a distance of 8 cm carry electric currents of 3A and 5 A. The distance of null point from the conductor carrying larger current when currents are flowing in the same direction is (in cm)
- Two long straight parallel conductors 10 cm apart, carry currents of 5A each in the same direction. Then the magnetic induction at a point mid-way between them is ____ (in T)
- Two long parallel wires are separated by a distance of 2 m. They carry a current of 1 A each in opposite direction. The magnetic induction at the midpoint of a straight line connecting these two wires is ____ (in μT)

LEVEL-II (ADVANCED)

Straight Objective Type Questions

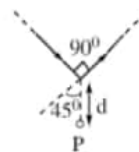
1. Find the magnetic field at P due to the arrangement shown

a) $\frac{\mu_0 i}{\sqrt{2} \pi d} \left(1 - \frac{1}{\sqrt{2}}\right) \otimes$

b) $\frac{2\mu_0 i}{\sqrt{2} \pi d} \otimes$

c) $\frac{\mu_0 i}{\sqrt{2} \pi d} \otimes$

d) $\frac{\mu_0 i}{\sqrt{2} \pi d} \left(1 + \frac{1}{\sqrt{2}}\right) \otimes$



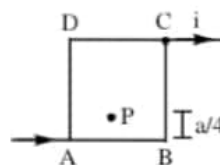
2. Figure shows a square loop of the edge a made of a uniform wire. A current i enters the loop at the point A and leaves it at the point C. Find the magnetic field at the point P which is on the perpendicular bisector of AB at a distance $a/4$ from it

a) $\frac{2\mu_0 i}{\pi a} \left(\frac{1}{\sqrt{5}} - \frac{1}{3\sqrt{13}}\right) \odot$

b) $\frac{2\mu_0 i}{a} \left(\frac{1}{\sqrt{5}} + \frac{1}{3\sqrt{13}}\right) \uparrow$

c) $\frac{\mu_0 i}{2a} \left(\frac{1}{\sqrt{5}} - \frac{1}{2\sqrt{13}}\right) \odot$

d) $\frac{2\mu_0 i}{\pi a^2} \left(\frac{1}{\sqrt{3}} + \frac{1}{3\sqrt{14}}\right) \otimes$



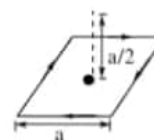
3. The magnetic field due to a current carrying square loop of side a at a point located symmetrically at a distance of $a/2$ from its centre (as shown is)

a) $\frac{\sqrt{2}\mu_0 i}{\sqrt{3}\pi a}$

b) $\frac{\mu_0 i}{\sqrt{6}\pi a}$

c) $\frac{2\mu_0 i}{\sqrt{3}\pi a}$

d) zero



4. A star shaped loop (with l = length of each section) carries current i . Magnetic field at the centroid of the loop is

a) $\frac{3\mu_0 i}{\pi l}$

b) $\frac{3\mu_0 i}{2\pi l}$

c) $(3 - \sqrt{3}) \frac{\mu_0 i}{\pi l}$

d) $(3 + \sqrt{3}) \frac{\mu_0 i}{\pi l}$



5. Three long straight conductors are arranged parallel to each other in the same plane and carry currents of 1 A, 2 A and 3 A all in the same direction. The distance between the first two conductors is " x " and the distance between the second and third conductors is " y ". If the middle conductor is in equilibrium, the ratio $x : y$ is

a) 1 : 3

b) 3 : 1

c) 1 : $\sqrt{3}$

d) $\sqrt{3} : 1$

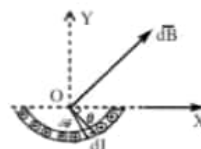
6. A current I flows in along straight wire with cross section having the form a half circular ring of radius R . The magnetic field induction at the point O is

a) $B = \frac{\mu_0 I}{4R}$

b) $B = \frac{\mu_0 I}{4\pi R}$

c) $B = \frac{\mu_0 I}{\pi^2 R}$

d) $B = \frac{\mu_0 I}{2\pi^2 R}$



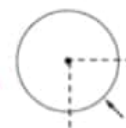
7. A long thin walled pipe of radius R carries a current I along its length. The current density is uniform over the circumference of the pipe. The magnetic field at the center of the pipe due to quarter portion of the pipe shown, is

a) $\frac{\mu_0 I \sqrt{2}}{4\pi^2 R}$

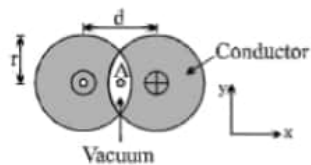
b) $\frac{\mu_0 I}{\pi^2 R}$

c) $\frac{2\mu_0 I \sqrt{2}}{\pi^2 R}$

d) None



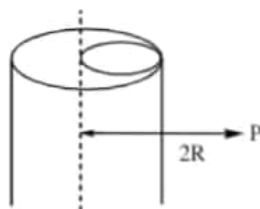
8. Two long conductors are arranged as shown below to form overlapping cylinders, each of radius r , whose centers are separated by a distance d . Current of density J flows into the plane of the page along the shaded part of one conductor and an equal current flows out of the plane of the page along the shaded portion of the other, as shown. What is the magnitude and direction of the magnetic field at point A?



- a) $\mu_0 Jd$ b) $\frac{\mu_0 Jd}{2}$ c) $\frac{\mu_0 Jd}{3}$ d) $\frac{\mu_0 Jd}{4}$

More than One correct answer Type Questions

9. A long straight wire carries a current along the x-axis. Consider the points A(0, 1, 0), B(0, 1, 1), C(1, 0, 1) and D(1, 1, 1). Which of the following pairs of points will have magnetic fields of the same magnitude?
- a) A and B b) A and C c) B and C d) B and D
10. A long, straight wire of radius R carries a current distributed uniformly over its cross-section. The magnitude of the magnetic field is
- a) maximum at the axis of the wire b) minimum at the axis of the wire
c) maximum at the surface of the wire d) minimum at the surface of the wire
11. An infinitely long current carrying thick cylindrical conductor, has a cylindrical cavity of radius $\frac{R}{2}$. The axis of the cylindrical cavity is at a distance $\frac{R}{2}$ from the axis of the cylindrical conductor. A current with current density J flows along the length of the conductor. The magnetic field at a point 'P' at a distance $2R$ from the axis of the cylinder is (point 'P' is on the plane containing both the axes)



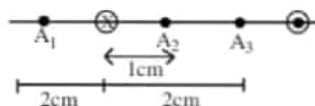
- a) magnetic field at P is $\frac{\mu_0 JR}{12}$
b) magnetic field at P is $\frac{\mu_0 JR}{6}$
c) magnetic field on the axis of the cavity is non-zero
d) magnetic field on the axis of the conductor is non-zero

Linked Comprehension Type Questions

Passage - I :

Figure shows two parallel wires separated by a distance of 4.0 cm and carrying equal currents of 10 A along opposite directions.

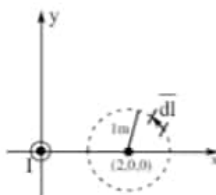
12. Find the magnitude of the magnetic field B at A_1 is



- a) $0.67 \times 10^{-4} \text{ T}$ b) $0.25 \times 10^{-4} \text{ T}$ c) $1.23 \times 10^{-9} \text{ T}$ d) $1.92 \times 10^{-6} \text{ T}$
13. Find the magnitude of the magnetic field B at A_2 is
a) $1.8 \times 10^{-6} \text{ T}$ b) $2.7 \times 10^{-4} \text{ T}$ c) $3.2 \times 10^{-3} \text{ T}$ d) $4.2 \times 10^{-4} \text{ T}$
14. Find the magnitude of the magnetic field B at A_3 is
a) $0.67 \times 10^{-4} \text{ T}$ b) $2.7 \times 10^{-4} \text{ T}$ c) $1.0 \times 10^{-5} \text{ T}$ d) $2.0 \times 10^{-4} \text{ T}$

Passage - II :

An infinitely long wire lying along z -axis carries a current I , flowing towards positive z -direction. There is no other current. Consider a circle in x - y plane with centre at $(2\text{ m}, 0, 0)$ and radius 1 meter. Divide the circle in small segments and let $d\vec{l}$ denote the length of a small segment in anticlockwise direction, as shown.



15. The path integral $\oint \vec{B} \cdot d\vec{l}$ of the total magnetic field $\vec{B}$ along the perimeter of the given circle is,
a) $\frac{\mu_0 I}{8}$ b) $\frac{\mu_0 I}{2}$ c) $\mu_0 I$ d) 0
16. Consider two points $A(3, 0, 0)$ and $B(2, 1, 0)$ on the given circle. The path integral $\int_A^B \vec{B} \cdot d\vec{l}$ of the total magnetic field $\vec{B}$ along the perimeter of the given circle from A to B is,
a) $\frac{\mu_0 I}{\pi} \tan^{-1} \frac{1}{2}$ b) $\frac{\mu_0 I}{2\pi} \tan^{-1} \frac{1}{2}$ c) $\frac{\mu_0 I}{2\pi} \sin^{-1} \frac{1}{2}$ d) 0
17. The maximum value of path integral $\oint \vec{B} \cdot d\vec{l}$ of the total magnetic field $\vec{B}$ along the perimeter of the given circle between any two points on the circle is
a) $\frac{\mu_0 I}{12}$ b) $\frac{\mu_0 I}{8}$ c) $\frac{\mu_0 I}{6}$ d) 0

Matrix Matching Type Questions

18. Column-II gives four situations in which three or four semiinfinite current carrying wires are placed in xy plane as shown. The magnitude of the direction of current is shown in each figure. Column-I gives statements regarding the x - and y -components of magnetic field at a point P whose coordinates are $(0, 0, d)$. Match the statements in Column-I with the corresponding figures in Column-II.

COLUMN - I

A) The x -component of magnetic field at point P is zero in

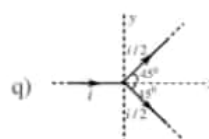
B) The z -component of magnetic field at point P is zero in

C) The magnitude of magnetic field at point P is $\frac{\mu_0 i}{4\pi d}$ in

D) The magnitude of magnetic field

at point P is greater than $\frac{\mu_0 i}{4\pi d}$ in

COLUMN - II



Integer Type Questions

19. An infinitely long solid conducting cylindrical rod of radius R is carrying a current distributed uniformly throughout the cross section. What is the ratio of magnetic fields at perpendicular distances $R/2$ and $2R$ from the axis of cylinder?
20. Two long straight parallel conductors 10 cm apart, carry current of 5A each in opposite directions. Then the magnetic induction at a point mid way between them is $x \times 10^{-5} \text{ T}$. What is the value of 'x'?

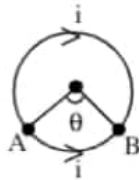
EXERCISE-II

(Magnetic field due to circular current loop)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. Magnetic field induction at centre of circular coil of radius 5 cm and carrying a current 0.9 A is (in S.I. units) (ϵ_0 = absolute permittivity of air S.I. units, velocity of light = $3 \times 10^8 \text{ m/s}$)
- 1) $\frac{1}{\epsilon_0 10^{16}}$ 2) $\frac{10^{16}}{\epsilon_0}$ 3) $\frac{\epsilon_0}{10^{16}}$ 4) $10^{16} \epsilon_0$
2. A circular coil of wire of radius 'r' has 'n' turns and carries a current 'I'. The magnetic induction (B) at a point on the axis of the coil at a distance $\sqrt{3} r$ from its centre is
- 1) $\frac{\mu_0 I n}{4 r}$ 2) $\frac{\mu_0 I n}{8 r}$ 3) $\frac{\mu_0 n I}{16 r}$ 4) $\frac{\mu_0 I n}{32 r}$

3. Two identical coils have a common centre and their planes are at right angles to each other and carry equal currents. If the magnitude of the induction field at the centre due to one of the coil is 'B', then the resultant magnetic induction field due to combination at their common centre is
 1) B 2) $\sqrt{2}B$ 3) $B/\sqrt{2}$ 4) 2B
4. A circular coil is made from a wire of length 2m. Its radius is $\frac{4}{\pi}$ cm. When a current of 1A passes through it, its magnetic moment is
 1) πAm^2 2) $\frac{4}{100\pi} \text{Am}^2$ 3) $\frac{16}{\pi} \text{Am}^2$ 4) $\frac{1}{\pi} \text{Am}^2$
5. A straight wire of length (π^2) meter is carrying a current of 2A and the magnetic field due to it is measured at a point distant 1 cm from it. If the wire is to be bent into a circle and is to carry the same current as before, the ratio of the magnetic field at its centre to that obtained in the first case would be
 1) 50 : 1 2) 1 : 50 3) 100 : 1 4) 1 : 100
6. Equal current 'i' flows in the two segments of a circular loop in the direction shown in fig. radius of the loop is 'a'. Magnetic field at the centre of the loop is,
 1) zero 2) $\left(\frac{\pi-\theta}{\pi}\right) \frac{\mu_0 i}{2a}$
 3) $\left(\frac{2\pi-\theta}{\pi}\right) \frac{\mu_0 i}{2a}$ 4) $\left(\frac{\theta}{2\pi}\right) \frac{\mu_0 i}{2a}$
- 
7. A circular coil of radius 2R is carrying current 'i'. The ratio of magnetic fields at the centre of the coil and at a point at a distance 6R from the centre of the coil on the axis of the coil is,
 1) 10 2) $10\sqrt{10}$ 3) $20\sqrt{5}$ 4) $20\sqrt{10}$
8. The field normal to the plane of a coil of 'n' turns and radius 'r' which carries a current 'i' is measured on the axis of the coil at a small distance 'h' from the centre of the coil. This is smaller than the field at the centre by the fraction.
 1) $\frac{3}{2} \frac{h^2}{r^2}$ 2) $\frac{2}{3} \frac{h^2}{r^2}$ 3) $\frac{3}{2} \frac{r^2}{h^2}$ 4) $\frac{2}{3} \frac{r^2}{h^3}$

Numerical Value Type Questions

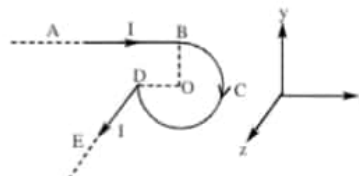
9. The electric current in a circular coil of two turns produced a magnetic induction of 0.2T at its centre. The coil is unwound and rewound in to a coil of four turns. The magnetic induction at the centre of the coil now is, in tesla (if the same current flows in the coil)
10. Two concentric circular coils A and B have radii 25 cm and 15 cm and carry currents 10 A and 15A respectively. A has 24 turns and B has 18 turns. The direction of currents are in opposite order. The magnetic induction at the common centre of the coils is $n\mu_0 T$, the value of n
11. The magnetic induction at the centre of a current carrying circular coil of radius 10 cm is $5\sqrt{5}$ times the magnetic induction at a point on its axis. The distance of the point from the centre of the coil in cm is

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. AB and DE are very long straight non-coplanar conductors at right angles. A conductor BCD in the form of $3/4^{\text{th}}$ of a circle of radius r connects the two straight conductors. If the circular coil is in the plane of AB (i.e., xy - plane) and normal to DE, then the magnetic induction at the centre O, due to current I is

- a) $\frac{\mu_0 I}{8\pi r} [3\pi + 4](-\hat{k})$ b) $\frac{\mu_0 I}{8\pi r} [(3\pi + 2)(-\hat{k}) + 2(\hat{j})]$
 c) $\frac{\mu_0 I}{8\pi r} [3\pi + 4](\hat{k})$ d) $\frac{\mu_0 I}{8\pi r} [(4 + 3\pi)(\hat{k}) + 2\hat{j}]$

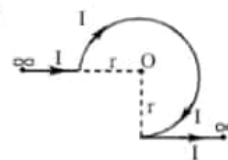


2. A charged non-conducting thin Disc of radius R is spinning with an angular speed ' ω '. Surface charge density of disc is ' σ ' then magnetic field at its centre is.

- a) $\frac{\mu_0 \sigma R}{2\omega}$ b) $\frac{\mu_0 \sigma \omega R}{2}$ c) $\frac{\mu_0 \sigma \omega R^2}{2}$ d) $\frac{\mu_0 \sigma \omega R}{2\pi}$

3. The magnetic induction at point 'O' in the following fig. will be

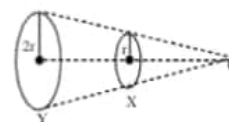
- a) $\frac{\mu_0 I}{4r} \left[\frac{3}{2} - \frac{1}{\pi} \right] \odot$ b) $\frac{\mu_0 I}{4r} \left[\frac{3}{2} - \frac{1}{\pi} \right] \otimes$
 c) $\frac{\mu_0 I}{4r} \left[\frac{3}{2} + \frac{1}{\pi} \right] \odot$ d) $\frac{\mu_0 I}{4r} \left[\frac{3}{2} + \frac{1}{\pi} \right] \otimes$



More than One correct answer Type Questions

4. Two circular coils X and Y having equal number of turns and carrying equal currents in same sense are placed co-axially, such that they subtend the same solid angle at O. Let B be the magnetic field at O due to smaller coil X, then

- a) Magnetic field at O due to larger coil = $2B$
 b) Magnetic field at O due to larger coil = $B/2$
 c) Total magnetic field at O is $3B/2$
 d) Total magnetic field at O is $\frac{\sqrt{5} B}{2}$



Linked Comprehension Type Questions

Passage :

There is plane spiral coil made on a thin insulated wire and has N turns Radii of inner and outer turns are a and b respectively



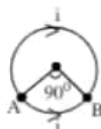
5. If a current I is passed through the spiral, the magnetic induction at the centre of the spiral will be
 a) $\frac{\mu_0 NI \log_e (b/a)}{2(b-a)}$ b) $\mu_0 \log_e \left(\frac{b}{a}\right)$ c) $\frac{1}{2} \mu_0 NI \log_e (b/a)$ d) $\frac{\mu_0 NI \log_e (b/a)}{(b-a)}$
6. The magnetic moment of this spiral is
 a) $\pi NI \log(b/a)$ b) $\pi NI(a^2 + b^2 + ab)$ c) $\frac{\pi NI \log(b/a)}{(b-a)}$ d) $\frac{1}{3} \pi NI(a^2 + b^2 + ab)$
7. If a uniform magnetic field is switched on perpendicular to plane of the coil, then
 a) Net force on the coil will be zero b) Net torque will be zero
 c) Net force on the coil is not zero d) None of these

Integer Type Questions

8. The magnetic field at the centre of a semicircle of radius 5cm of passing a current i through it $2\pi \times 10^{-6} \text{ T}$. Then find the value of $2i$.
9. Two concentric coplanar semicircular conductors from part of two current loops as shown in the figure. If their radii are 11 cm and 4 cm. If the magnetic induction at the centre O is $x \times 10^{-3} \text{ Wb/m}^2$. Find the value of x .



10. Equal currents of 2A flows in the two segments of a circular loop as shown in fig. radius of circular loop is 10cm. Magnetic field at the centre of the loop 'O' is $n\pi \times 10^{-6}$. Where 'n' is



11. Magnetic field at two points on the axis of a circular coil at a distance of 0.05 m and 0.2 m from the centre are in the ratio 8 : 1. The radius of the coil is $x \times 10^{-1} \text{ m}$. What is the value of 'x' ?

EXERCISE-III (Ampere's Law, Solenoid and Toroid)

LEVEL-I (MAIN)

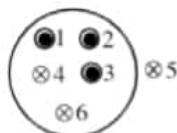
Straight Objective Type Questions

1. A current I flows along a round loop. Find $\int \vec{B} \cdot d\vec{r}$ along the axis of the loop within the range from $-\infty$ to $+\infty$.
 1) $\mu_0 I / 2\pi$ 2) $2I / \mu_0$ 3) μ_0 / I 4) $\mu_0 I$
2. If current density in a conducting wire is proportional to the distance r from the axis of the conductor. Find magnetic field at the position $r < R$, where R is the radius of cross section of the conductor. (i is the current in conducting wire)
 1) $\frac{\mu_0 i r^2}{2\pi R^3}$ 2) $\frac{\mu_0 i r^2}{4\pi R^3}$ 3) $\frac{\mu_0 i r^2}{2\pi R^2}$ 4) $\frac{\mu_0 i r^2}{2\pi R^4}$

3. A current carrying wire carries a current of $2A$, lies along the z -axis which is out of page, another wire carrying current of $4A$ in same direction lies parallel to the first. Then around which loop linking both the wires $\int \vec{B} \cdot d\vec{r}$ is zero.



4. Six wires of current $I_1 = 1A$, $I_2 = 2A$, $I_3 = 3A$, $I_4 = 1A$, $I_5 = 5A$ and $I_6 = 4A$ cut the page perpendicular at the points 1, 2, 3, 4, 5 and 6 respectively as shown in the figure. Find value of the integral $\oint \vec{B} \cdot d\vec{l}$ around the closed path.



- 1) μ_0 2) $2\mu_0$ 3) $3\mu_0$ 4) $4\mu_0$

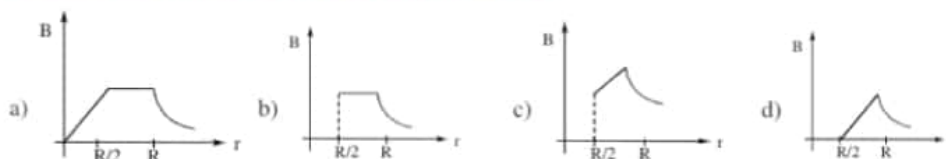
Numerical Value Type Questions

5. An infinite wire carrying a current i is passing through the centre of a circular loop of radius R . The length of wire makes an angle 60° with the plane of the loop. $\int \vec{B} \cdot d\vec{l} = \frac{\sqrt{x}}{2} \mu_0 i$ when evaluated along the loop. Find x .
6. The length of a solenoid is $0.1m$ and its diameter is very small. A wire is wound over in two layers. The number of turns in the inner layer is 50 and that on the outer layer is 40 . The strength of current flowing in two layers in the opposite direction is 3 ampere. The magnetic induction in the middle of the solenoid will be $N \times 10^{-4} T$, where N is
7. A toroidal solenoid has 3000 turns and a mean radius of $10cm$. It has a soft iron core of relative permeability 2000 . Find the magnetic field in the core when a current of $1.0A$ is passed through the solenoid. (in T)
8. A long solenoid has 200 turns per cm and carries a current i , the magnetic field at its centre is $6.28 \times 10^{-2} Wb/m^2$, another long solenoid has 100 turns per cm and it carries a current $i/3$. The value of the magnetic field at its centre is $______ n \times 10^{-2} wb/m^2$, the value of n

LEVEL-II (ADVANCED)

Straight Objective Type Questions

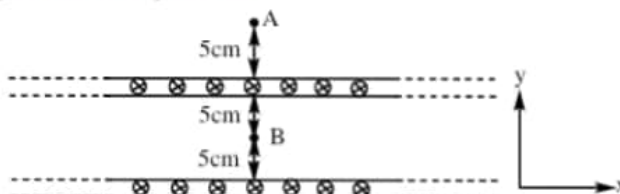
1. An infinitely long hollow conducting cylinder with inner radius $R/2$ and outer radius R carries a uniform current density along its length. The magnitude of the magnetic field, $|B|$ as a function of radial distance r from the axis is best represented by



2. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii a and b , respectively. When a current I passes through the coil, then magnetic field at the centre is

a) $\frac{\mu_0 NI}{b}$ b) $\frac{2\mu_0 NI}{a}$ c) $\frac{\mu_0 NI}{2(b-a)} \ln\left(\frac{a}{b}\right)$ d) $\frac{\mu_0 NI}{2(b-a)} \ln\left(\frac{b}{a}\right)$

3. Figure shows cross section of two large parallel metal sheets carrying electric currents along their surfaces. The current in each sheet is $\frac{10}{\pi}$ A/m along the sheet. Consider two points A and B, as shown in the figure with their positions.



- a) Magnetic field at A is $4\mu\text{T}$ along x-direction
 b) Magnetic field at A is $4\mu\text{T}$ along negative x-direction
 c) Magnetic field at B is $4\mu\text{T}$ along x-direction
 d) Magnetic field at B is $2\mu\text{T}$ along x-direction.

Linked Comprehension Type Questions

Passage - I :

Ampere's law gives a method to calculate the magnetic field due to given current distribution.

The circulation $\oint \vec{B} \cdot d\vec{l}$ of the resultant magnetic field along a closed, plane curve is equal to μ_0 times the total current crossing the area bounded by the closed curve provided the electric field inside the loop remains constant.



Thus $\oint \vec{B} \cdot d\vec{l} = \mu_0(i_1 - i_2)$

The contribution of current i_3 to the magnetic field cancel out because the integration is made around the full loop.

4. Consider a co-axial cable which consists of an inner wire of radius a surrounded by an outer shell of inner and outer radii b and c respectively. The inner wire carries a current i and outershell carries an equal and opposite current. The magnetic field at a distance x from axis for $x < a$ is

a) zero b) $\frac{\mu_0 i}{2\pi x}$ c) $\frac{\mu_0 i x^2}{2\pi a^3}$ d) $\frac{\mu_0 i x}{2\pi a^2}$

5. Refer to above question, the magnetic field a distance x from the axis where $a < x < b$ is

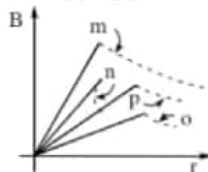
a) $\frac{\mu_0 i}{2\pi x}$ b) $\frac{\mu_0 i}{2\pi(b-a)}$ c) $\frac{\mu_0 i x}{2\pi(b^2 - a^2)}$ d) zero

6. The magnetic field at a distance x from the axis where $b < x < c$ is

a) $\frac{\mu_0 i (c^2 - b^2)}{2\pi x (c^2 - a^2)}$ b) $\frac{\mu_0 i (c^2 - x^2)}{2\pi x (c^2 - a^2)}$ c) $\frac{\mu_0 i (c^2 - x^2)}{2\pi x (c^2 - b^2)}$ d) zero

Passage - II :

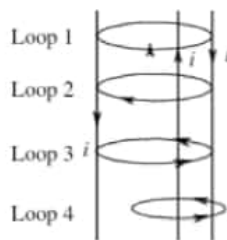
Curves in the graph shown as function of radial distance r , the magnitude B of the magnetic field inside and outside four long wires m , n , o and p carrying currents that are uniformly distributed across the cross-section of the wires. Overlapping portions of the plots are indicated by double labels.



7. Which wire has the greatest radius ?
 a) m b) n c) o d) p
8. Which wire has the greatest magnitude of the magnetic field on the surface ?
 a) m b) n c) o d) p
9. The current density in the wire m is
 a) greater than in wire n b) less than in wire n
 c) equal to that in wire n d) not comparable due to lack of information

Matrix Matching Type Questions

10. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown in the figure. The direction of $d\mathbf{l}$ is shown in the figure.



Then, match the Column I with Column II and select the correct option from the codes given below.

COLUMN I

- A) Along closed loop 1
 B) Along closed loop 2
 C) Along closed loop 3
 D) Along closed loop 4

COLUMN II

- p) $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$
 q) $\oint \mathbf{B} \cdot d\mathbf{l} = -\mu_0 I$
 r) $\oint \mathbf{B} \cdot d\mathbf{l} = 0$
 s) net work done by the magnetic force to move a unit charge along the loop is zero
 t) $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (2I)$

	A	B	C	D
a)	q,s	p,s	r,s	r,s
c)	p	q	r,s	p,s

	A	B	C	D
b)	p,q	p,s	q,s	s,s
d)	q,r	t,s	p,q	s,t

Integer Type Questions

11. A long conducting cylinder of radius R carries a current I . The current density J is a function of radius r as $J = br$, where b is a constant. Magnetic field at a distance $r > R$ measured from the axis is $B = \frac{\mu_0 b R^3}{Nr_2}$, then find the value of N .

EXERCISE-IV*(Motion of the charged particle in the magnetic field)***LEVEL-I (MAIN)**Straight Objective Type Questions

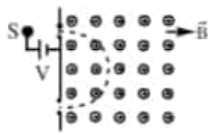
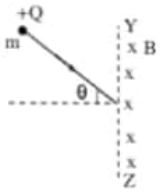
- A proton enters a magnetic field of flux density 1.5 T with a velocity of $20 \times 10^7 \text{ ms}^{-1}$ at an angle of 30° with the field. The force on the proton is [$e_p = 1.6 \times 10^{-19} \text{ C}$]
 1) $2.4 \times 10^{-10} \text{ N}$ 2) $2.4 \times 10^{-11} \text{ N}$ 3) $3 \times 10^{-5} \text{ N}$ 4) $3 \times 10^{-4} \text{ N}$
- A 2 MeV proton is moving perpendicular to a uniform magnetic field of 2.5 T the force on the proton is (mass of proton = $1.6 \times 10^{-27} \text{ kg}$)
 1) $10 \times 10^{-12} \text{ N}$ 2) $8 \times 10^{-11} \text{ N}$ 3) $2.5 \times 10^{-10} \text{ N}$ 4) $8 \times 10^{-12} \text{ N}$
- An electron moves with speed $2 \times 10^5 \text{ ms}^{-1}$ along the positive x -direction in the presence of a magnetic induction $\vec{B} = \hat{i} + 4\hat{j} - 3\hat{k}$ tesla. The magnitude of the force experienced by the electron in newton is (charge on the electron = $1.6 \times 10^{-19} \text{ C}$)
 1) 1.18×10^{-13} 2) 1.28×10^{-13} 3) 1.6×10^{-13} 4) 1.72×10^{-13}
- An electron experiences a force $(4.0\hat{i} + 3.0\hat{j}) \times 10^{-13} \text{ N}$ in a uniform magnetic field when its velocity is $2.5\hat{k} \times 10^7 \text{ m/s}$. When the velocity is redirected and becomes $(1.5\hat{i} - 2.0\hat{j}) \times 10^7 \text{ m/s}$, the magnetic force of the electron is zero. The magnetic field vector $\vec{B}$ is
 1) $-0.075\hat{i} + 0.1\hat{j}$ 2) $0.1\hat{i} + 0.075\hat{j}$ 3) $0.075\hat{i} - 0.1\hat{j} + \hat{k}$ 4) $0.075\hat{i} - 0.1\hat{j}$

Numerical Value Type Questions

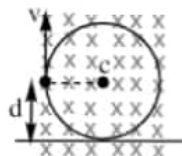
- A charged particle of charge $4mc$ enters a uniform magnetic field of induction $\vec{B} = 4\hat{i} + y\hat{j} + z\hat{k}$ tesla with a velocity $\vec{v} = 2\hat{i} + 3\hat{j} - 6\hat{k}$. If the particle continues to move undeviated, then the strength of the magnetic field induction (B) is ____ tesla
- A beam 30 mev α -particle is to be obtained from a 50 cm cyclotron. The strength of magnetic field required to be applied will be (in T)
- A clycotron is used to obtain 2 Mev protons, if the frequency is 5 MHz and the potential is 20 kv . The magnetic field necessary for resonance is (in T)

LEVEL-II (ADVANCED)Straight Objective Type Questions

- A uniform magnetic field $\vec{B} = B_0\hat{j}$ exists in a space. A particle of mass m and charge q is projected towards negative x -axis with speed v from the a point $(d, 0, 0)$. The maximum value v for which the particle does not hit y - z plane is
 a) $\frac{2Bq}{dm}$ b) $\frac{B_0qd}{m}$ c) $\frac{Bq}{2dm}$ d) $\frac{Bqd}{2m}$

2. Two protons move parallel to each other, keeping distance r between them, both moving with same velocity $\vec{V}$. Then the ratio of the electric and magnetic force of interaction between them is
 a) c^2/V^2 b) $2c^2/V^2$ c) $c^2/2V^2$ d) None
3. A charged particle of specific charge α is released from origin at time $t = 0$ with velocity $\vec{V} = V_0\hat{i} + V_0\hat{j}$ in magnetic field $\vec{B} = B_0\hat{i}$. The co-ordinates of the particle at time $t = \frac{\pi}{B_0\alpha}$ are (specific charge $\alpha = q/m$)
 a) $\left(\frac{V_0}{2B_0\alpha}, \frac{\sqrt{2}V_0}{\alpha B_0}, \frac{-V_0}{B_0\alpha}\right)$ b) $\left(\frac{-V_0}{2B_0\alpha}, 0, 0\right)$ c) $\left(0, \frac{2V_0}{B_0\alpha}, \frac{V_0\pi}{2B_0\alpha}\right)$ d) $\left(\frac{V_0\pi}{B_0\alpha}, 0, \frac{-2V_0}{B_0\alpha}\right)$
4. A mass spectrometer is a device which select particle of equal mass. An ion with electric charge $q > 0$ and mass m starts at rest from a source S and is accelerated through a potential difference V . It passes through a hole into a region of constant magnetic field $\vec{B}$ perpendicular to the plane of the paper as shown in the figure. The particle is deflected by the magnetic field and emerges through the bottom hole at a distance d from the top hole. The mass of the particle is
 a) $\frac{qBd}{8V}$ b) $\frac{qB^2d^2}{4V}$
 c) $\frac{qB^2d^2}{8V}$ d) $\frac{qBd}{2V}$
- 
5. A particle of specific charge α (charge per unit mass) is released at time $t = 0$ from origin with an initial velocity of $\vec{v} = v_0\hat{i}$ in a uniform magnetic field $\vec{B} = -B_0\hat{k}$. Find position of particle at any time t .
 a) $\frac{v_0}{B_0\alpha} [\sin(B_0\alpha t)\hat{i} - \{1 + \cos(B_0\alpha t)\}\hat{j}]$ b) $\frac{v_0}{B_0\alpha} [\sin(B_0\alpha t)\hat{i} + \{1 - \cos(B_0\alpha t)\}\hat{j}]$
 c) $\frac{v_0}{B_0\alpha} [\cos B_0\alpha t + \{1 - \sin(B_0\alpha t)\}\hat{j}]$ d) $\frac{v_0}{B_0\alpha} [\sin\alpha t\hat{i} - \{1 + \cos(\alpha t)\}\hat{j}]$
6. A particle with charge $+Q$ and mass m enters a magnetic field of magnitude B , existing only to the right of the boundary YZ . The direction of the motion of the particle is perpendicular to the direction of B . Let $T = 2\pi \frac{m}{QB}$. The time spent by the particle in the field will be
 a) $T\theta$ b) $2T\theta$
 c) $T\left(\frac{\pi + 2\theta}{2\pi}\right)$
 d) $T\left(\frac{\pi - 2\theta}{2\pi}\right)$
- 
7. In the previous question, if the particle has $-Q$ charge, the time spent by the particle in the field will be
 a) $T\theta$ b) $2T\theta$ c) $T\left(\frac{\pi + 2\theta}{2\pi}\right)$ d) $T\left(\frac{\pi - 2\theta}{2\pi}\right)$

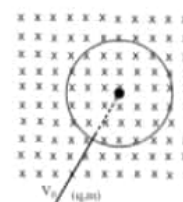
8. A plane metallic sheet is placed with its face parallel to lines of magnetic induction B of a uniform field. A particle of mass m and charge q is projected with a velocity v from a distance ' d ' from the plane normal to the lines of induction. Then the maximum velocity of projection for which the particle does not hit the plane is,



- a) $\frac{2Bqd}{m}$ b) $\frac{Bqd}{m}$ c) $\frac{Bqd}{2m}$ d) $\frac{Bqm}{d}$

More than One correct answer Type Questions

9. A beam of protons with a velocity of $4 \times 10^5 \text{ ms}^{-1}$ enters a uniform magnetic field of 0.3 T . The velocity makes an angle of 60° with the magnetic field.
- The radius of the helical path taken by the proton is 1.2 cm
 - The radius of the helical path taken by the proton is 12 cm
 - The pitch of the helix is 4.4 mm
 - The pitch of the helix is 4.2 cm
10. Two ions have equal masses but one is singly ionized and the other is doubly ionized. They are projected from the same place in a uniform magnetic field with the same velocity perpendicular to the field.
- Both ions will move along circles of equal radii
 - The circle described by the singly-ionized charge will have a radius double that of the other circle
 - The two circles do not touch each other
 - The two circles touch each other
11. A non-conducting cylindrical shell of radius R is placed in uniform magnetic field. The axis of the cylinder is parallel to the direction of magnetic field. A hole is drilled in cylinder and a particle having charge q and mass m is projected with velocity v_0 perpendicular to magnetic field and directed towards the axis of cylinder as shown in diagram. The particle collides elastically with inner wall of the cylinder and rebounds.
- The minimum number of collisions after which it will emerge is two.
 - The maximum speed v_0 for which it can come out is $v_0 = \left(\frac{qBR}{m} \right) \sqrt{3}$.
 - The minimum time in which it can come out will be $\left(\frac{\pi m}{qB} \right)$
 - If the particle is projected with a velocity $\frac{V_0}{3}$ {where $V_0 = \left(\frac{qBR}{m} \right) \sqrt{3}$ } it will come out after five collisions.



12. Which of the following statement is/are correct :

- A charged particle enters a region of uniform magnetic field at an angle 85° to magnetic lines of force. The path of the particle is a circle.
- An electron and proton are moving with the same kinetic energy along the same direction. When they pass through uniform magnetic field perpendicular to their direction of motion, they describe circular path.
- There is no change in the energy of a charged particle moving in a magnetic field although magnetic force acts on it.
- Two electrons enter with the same speed but in opposite direction in a uniform transverse magnetic field. Then the two describe circle of the same radius and these move in the same direction.

Linked Comprehension Type Questions

Passage :

13. Figure shows a circular wire loop of radius a carrying a current i , in anti clockwise direction placed in perpendicular magnetic field B consider a small part dl of the wire find the force on the part of the wire exerted by the magnetic field



- $iBdl$ (towards centre)
 - iB/dl (towards centre)
 - idl (towards centre)
 - $\mu_0 idlB$ towards centre
14. Find the tension in the wire if a current i flows in clockwise direction
- $iBdl$
 - iBa
 - $\mu_0 iBdl$
 - $iBdl a$
15. Suppose that the radius of cross-section of the wire used is r . Find the decrease in the radius of the loop if the magnetic field is switched off. The Young's modulus of the material of the wire is Y .
- $\frac{Biadl}{Y}$
 - $\frac{iaB}{2\pi r^2 Y}$
 - $\frac{ia^2 B}{\pi r^2 Y}$
 - $\frac{a^2 B}{4\pi r^2 Y}$

Integer Type Questions

- A beam of electrons is accelerated through a potential difference 'V'. It is then passed normally through a uniform magnetic field where it moves in a circle of radius 'r'. It would have moved in a circle of radius $2r$ if it were initially accelerated through a potential difference of kV , then $k =$
- An experimenter's diary reads as follows: "a charged particle is projected in a magnetic field of $(7.0\hat{i} - 3.0\hat{j}) \times 10^{-3} \text{ T}$. The acceleration of the particle is found to be $(\Omega\hat{i} + 7.0\hat{j}) \times 10^{-6} \text{ m/s}^2$. The number to the left of $\hat{i}$ in the last expression was not readable. What can this number be ?

EXERCISE-V

(Motion of charged particle in combined electric and magnetic field)

LEVEL-I (MAIN)

Straight Objective Type Questions

- A beam of protons enters a uniform magnetic field of 0.3 T with a velocity of $4 \times 10^5 \text{ m/sec}$ in a direction making an angle of 60° with the direction of magnetic field, the pitch of the helix will be
 - 4.7 m
 - 0.47 m
 - 0.047 m
 - 0.0047 m

2. An electron beam passes through a magnetic field of $2 \times 10^{-3} \text{ Wb/m}^2$ and electric field of $3.2 \times 10^4 \text{ V/m}$, both acting simultaneously ($\vec{E} \perp \vec{B} \perp \vec{V}$) if the path of electrons remains undeflected calculate the speed of the electron. If the electric field is removed, what will be the radius of the electron path. [mass of electron = $9.1 \times 10^{-31} \text{ kg}$]
- 1) $16 \times 10^6 \text{ m/s}$ $\frac{20}{91} \text{ cm}$ 2) $1.6 \times 10^5 \text{ m/s}$ $\frac{91}{20} \text{ cm}$ 3) $16 \times 10^6 \text{ m/s}$ $\frac{91}{20} \text{ cm}$ 4) $1.6 \times 10^5 \text{ m/s}$ $\frac{20}{91} \text{ cm}$
3. A particle moves in a circle of radius 1.0 cm under the action of a magnetic field of 0.40 T. An electric field of 200 V/m makes the path straight. Find the charge/mass ratio of the particle.
- 1) $\frac{5}{4} \times 10^5 \text{ C/kg}$ 2) $\frac{4}{5} \times 10^5 \text{ C/kg}$ 3) $\frac{5}{4} \times 10^6 \text{ C/kg}$ 4) $\frac{4}{5} \times 10^6 \text{ C/kg}$
4. A proton goes undeflected in a crossed electric and magnetic field (The fields are perpendicular to each other) at a speed of 10^5 m/s . The velocity is perpendicular to both the fields. When the electric field is switched off, the proton moves along a circle of radius 2 cm. Find the magnitudes of the electric and the magnetic fields. Take the mass of the proton = $1.6 \times 10^{-27} \text{ kg}$.
- 1) $5 \times 10^{-3} \text{ N/C}$, $5 \times 10^{-2} \text{ T}$ 2) $5 \times 10^3 \text{ N/C}$, $5 \times 10^{-2} \text{ T}$
 3) $5 \times 10^3 \text{ N/C}$, $5 \times 10^2 \text{ T}$ 4) $5 \times 10^2 \text{ N/C}$, $5 \times 10^{-3} \text{ T}$
5. A particle having mass m and charge q is released from the origin in a region in which electric field and magnetic field are given by $\vec{B} = +B_0 \hat{j}$ and $\vec{E} = +E_0 \hat{i}$ find the speed of the particle as a function of x co ordinate.
- 1) $\sqrt{\frac{2qE_0 x}{m}}$ 2) $\sqrt{\frac{qE_0 x}{m}}$ 3) $\sqrt{\frac{qE_0 x}{2m}}$ 4) $\sqrt{\frac{3qE_0 x}{m}}$

Numerical Value Type Questions

6. A particle having a charge $q = 1 \text{ C}$ and mass $m = 0.5 \text{ kg}$ is projected on a rough horizontal plane (x, y) from a point (15m, 0, 0) with a initial velocity $\vec{v} = 20 \hat{j}$. In space an uniform electric field $\vec{E} = 25(-\hat{k})$ and magnetic field $\vec{B} = 10(-\hat{k})$. The acceleration due to gravity $g = 10 \text{ msec}^{-2}$ and coefficient of friction $\mu = 0.5$. The particle moves on a spiral path and reaches to the origin. The time taken by particle to reach origin is found $y \times 10 \text{ sec}$. Find y .
7. A non-relativistic proton beam passes without deviation in a uniform manner through a region of space where there are uniform transverse mutually perpendicular electric and magnetic fields with $E = 120 \text{ kV/m}$ and $B = 25 \text{ mT}$. Then the beam strikes a grounded target which helps the protons to come to rest. The force with which the beam acts on the target if the beam current is equal to $I = 8 \text{ mA}$ is $F \mu \text{N}$, Find the value of F ? (Specific charge of proton = $9.6 \times 10^7 \text{ C/kg}$)

LEVEL-II (ADVANCED)

Straight Objective Type Questions

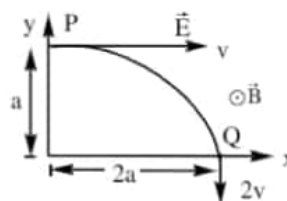
1. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0 \hat{i}$ and $\vec{B} = B_0 \hat{i}$ with velocity $\vec{v} = v_0 \hat{j}$. The speed of the particle will become $2v_0$ after a time
- a) $t = \frac{2mv_0}{qE}$ b) $t = \frac{2Bq}{mv_0}$ c) $t = \frac{\sqrt{3}Bq}{mv_0}$ d) $t = \frac{\sqrt{3}mv_0}{qE}$

2. A particle of specific charge (q/m) is projected from the origin of coordinates with initial velocity $[u\hat{i} - v\hat{j}]$. Uniform electric magnetic fields exist in the region along the $+y$ direction, of magnitude E and B . The particle will definitely return to the origin once if
- a) $[vB/2\pi E]$ is an integer b) $(u^2 + v^2)^{1/2} [B/\pi E]$ is an integer
- c) $[vB/\pi E]$ is an integer d) $[uB/\pi E]$ is an integer

More than One correct answer Type Questions

3. A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E\hat{i}$ and uniform magnetic field $B\hat{k}$ follows a trajectory from P to Q as shown in figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$. Which of the following statement(s) is/are correct ?

- a) $E = \frac{3}{4} \left(\frac{mv^2}{qa} \right)$
- b) Rate of work done by electric field at P $= \frac{3}{4} \left(\frac{mv^3}{a} \right)$
- c) Rate of work done by electric field at P is zero
- d) Rate of work done by both the fields at Q is zero



4. Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a , b and c be the ratio of their time periods, radii and pitches of the helical paths than
- a) $abc = 1$ b) $abc > 1$ c) $abc < 1$ d) $a = bc$

Linked Comprehension Type Questions

Passage - I :

A charged particle of mass m and charge q is allowed to enter in a region of space in which uniform electric and magnetic field can be created. The subsequent path of the particle in the region will depend upon initial value of velocity and direction of the field

5. If $E = 0$, $B = B_0\hat{k}$ and $V = V_0\hat{i} + V_0\hat{j}$ then the path of the particle is
- a) Circle of radius $\frac{mv_0}{qB}$ in xy plane b) helical with uniform pitch $\frac{2\pi mv_0}{qB}$
- c) a circle of radius $\frac{\sqrt{2}mv_0}{qB}$ in xy plane d) helical with non-uniform pitch
6. If $E = E_0\hat{k}$, $B = B_0\hat{k}$ and $V = V_0\hat{i}$ then the path of the particle will be
- a) the path of the particle is a helix with a non uniform pitch
- b) the path of the particle is helix with uniform pitch
- c) the maximum velocity of the particle in the subsequent motion is v_0
- d) the velocity of the particle in the subsequent motion decreases uniformly
7. If $E = E_0\hat{k}$, $B = B_0\hat{k}$ and $V = V_0\hat{i} + V_0\hat{k}$ then
- a) a helix with increasing pitch and axis parallel to z-axis
- b) a cycloid in the x-y plane c) a cycloid the y-z plane
- d) a cycloid in the x-z plane

Passage - II :

A particle with specific charge α (charge per unit mass) moves in the region of space where mutually perpendicular electric field $\vec{E} = E_0 \hat{j}$ and magnetic field $\vec{B} = B_0 \hat{k}$ are present. At time $t = 0$ particle is located at origin with zero initial velocity.

8. Find acceleration of the particle at $t = 0$

a) $\frac{E_0 q}{m} \hat{j}$ b) $\frac{E_0 q}{2m} \hat{i}$ c) $\frac{E_0 q}{m} \hat{k}$ d) $\frac{E_0}{mq} \hat{i}$

9. Find x-coordinate of the particle at time 't'.

a) $x = \frac{E_0}{B_0 \omega} \sin \omega t$ b) $x = \frac{E_0}{B_0 \omega} (\omega t - \sin \omega t)$ c) $x = \frac{E_0}{B_0 \omega} (\omega t + \sin \omega t)$ d) $x = \frac{E_0}{B_0 \omega} \cos \omega t$

10. Find y-coordinate of the particle at time 't'

a) $y = \frac{E_0}{B_0 \omega} (1 + \cos \omega t)$ b) $y = \frac{E_0}{B_0 \omega} \sin \omega t$ c) $y = \frac{E_0}{B_0 \omega} (1 - \cos \omega t)$ d) $y = \frac{E_0}{B_0 \omega} \cos \omega t$

Matrix Matching Type Questions

11. For the path of a charged particle match the following.

COLUMN - I

- A) in uniform electric field
B) in uniform magnetic field
C) in uniform electric and magnetic field which are directed parallel to each other
D) in uniform electric and magnetic field which are perpendicular to each other

COLUMN - II

- p) Straight line
q) Parabola
r) Circle
s) Helix

Integer Type Questions

12. A uniform charged ring of radius 10 cm rotates at a frequency of 10^4 rps about its axis the ratio of energy density of electric field to the energy density of the magnetic field at a point on the axis at distance 20 cm from the centre is $x \times 10^9$ approximately. Find x.
13. A current I flows along the length of a thin walled, long metallic hollow cylinder of radius R , distributed uniformly on its surface. If the pressure on the wall is $P = \left(\frac{\mu_0 I^2}{\pi^2 R^2} \right) \times \left(\frac{1}{x} \right)$. Find the value of x.

EXERCISE-VI

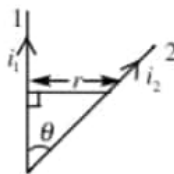
(Magnetic force between two parallel conductors)

LEVEL-I (MAIN)**Straight Objective Type Questions**

1. Two long parallel copper wires carrying current in the opposite direction of 5A each. If the wires are separated by a distance of 0.5m, then the force between the two wires is
- 1) 10^{-5} N/m attractive 2) 10^{-5} N/m repulsive
3) 2×10^{-6} N/m attractive 4) 2×10^{-5} N/m repulsive

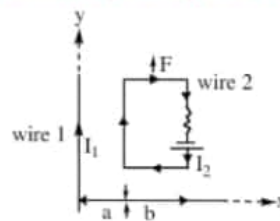
2. Wires 1 and 2 carrying currents i_1 and i_2 respectively are inclined at an angle θ to each other. What is the force on a small element dl of wire 2 at a distance r from wire 1 due to magnetic field of wire 1?

- 1) $\frac{\mu_0}{4\pi r} i_1 i_2 dl [1 + \cos \theta]$ 2) $\frac{\mu_0}{2\pi r} i_1 i_2 dl \sin \theta$
 3) $\frac{\mu_0}{2\pi r} i_1 i_2 dl \cos \theta$ 4) $\frac{\mu_0}{4\pi r} i_1 i_2 dl \sin \theta$



3. Wire-1 in fig is oriented along the Y-axis and carries a steady current I_1 . A rectangular circuit located to the right of the wire carries a current I_2 . Find the force experienced by the top of the horizontal wire (wire-2) of the rectangular circuit.

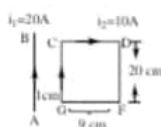
- 1) $\left[\frac{\mu_0 I_1 I_2}{2\pi} \ln \left(1 + \frac{b}{a} \right) \right] \hat{j}$ 2) $\left[\frac{\mu_0 I_1 I_2}{2\pi} \ln \left(1 + \frac{b}{a} \right) (-\hat{j}) \right]$
 3) $\left[\frac{\mu_0 I_1 I_2}{2\pi} \ln \left(1 + \frac{b}{a} \right) (-\hat{i}) \right]$ 4) $\left[\frac{\mu_0 I_1 I_2}{2\pi} \ln \left(1 + \frac{b}{a} \right) (\hat{i}) \right]$



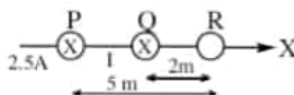
4. Two parallel horizontal conductors are suspended by light vertical threads each of length 75cm. Each conductor has a mass of 40gm per m. When no current flows through them, they are 0.5 cm apart. When same current flows through each conductor the separation is 1.5 cm. The value and direction of current is
- 1) 14 A in same direction 2) 14 A in opposite direction
 3) 196 A in same direction 4) 196 A in opposite direction
5. A current of 3A is flowing in a linear conductor having a length of 40 cm. The conductor is placed in a magnetic field of strength 500 Gauss and makes an angle of 30° with the direction of the field. It experiences a force of magnitude
- 1) 3×10^4 N 2) 3×10^2 N 3) 3×10^{-2} N 4) 3×10^{-4} N

Numerical Value Type Questions

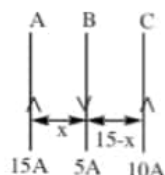
6. The force exerted by the conductor AB on the loop CDFG shown in figure is ($N \times 10^{-4}$ N attraction) the value of n is



7. Two long parallel wires carrying currents 2.5A and I amp in the same direction (directed into plane) are held at P and Q respectively as shown. The points P and Q are located at 5 m and 2m respectively from a collinear point R. An electron moving with a velocity of 4×10^5 m/s along positive x axis experiences a force of 3.2×10^{-20} N at the point 'R'. The value of I is (in A)



8. Three long straight and parallel wires carrying currents are arranged as shown in figure. The wire 'B' which carries a current of 5A is so placed that it experiences no force. The distance of wire 'B' from A is then (in cm)



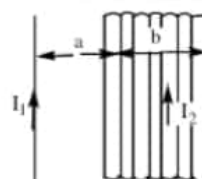
LEVEL-II (ADVANCED)

Straight Objective Type Questions

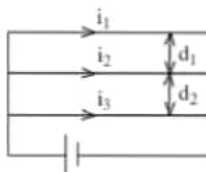
1. Two long thin parallel conductors of the shape shown in fig carry direct currents I_1 and I_2 . The separation between the conductors is a , the width of the right-hand conductor is equal to b . With both conductors lying in one plane, find the magnetic interaction force between them reduced to a unit of their length.

a) $\frac{\mu_0}{4\pi} \frac{I_1 I_2}{a} \ln(1+a/b)$ b) $\frac{\mu_0}{4\pi} \frac{2I_1 I_2}{b} \ln(1+b/a)$

c) $\frac{\mu_0}{4\pi} \frac{I_1 I_2}{b} \ln(1-a/b)$ d) $\frac{\mu_0}{4\pi} \frac{2I_1 I_2}{b} \ln(1-b/a)$

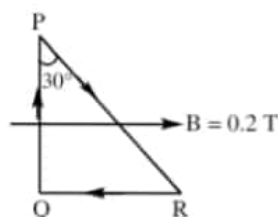


2. Three long wires of resistances in the ratio 3:4:5 are connected in parallel to each other as shown in figure. If net force on middle wire is zero then $\frac{d_1}{d_2}$ will be



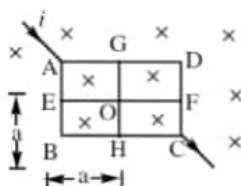
- a) 9.25 b) 5:3 c) $\sqrt{5}:\sqrt{3}$ d) 1:1

3. A wire bent in the form a right angled triangle PQR, carries a current I A. It is placed in a region of a uniform magnetic field $B = 0.2$ T. If $PR = 1$ m, the net force on the wire is



- a) 1.73 N b) 3.46N c) 2.732N d) Zero

4. The figure shows a uniform conducting structure which carries current i with each small square has side a . The structure is kept in a uniform magnetic field B . Then the magnetic force on the structure will be



- a) $2\sqrt{2} iBa$ b) $\sqrt{2} iBa$ c) $2 iBa$ d) iBa
5. A straight current carrying conductor is placed in such a way that the current in the conductor flows in the direction out of the plane of the paper. The conductor is placed between two poles of two magnets, as shown. The conductor will experience a force in the direction towards



- a) P b) Q c) R d) S

EXERCISE-VII

(Torque on current loop, MCG)

LEVEL-I (MAIN)

Straight Objective Type Questions

- A rectangular coil of wire of 100 turns and $10 \times 15\text{cm}^2$ size carrying a current of 2Amp. is in a magnetic field of induction $2 \times 10^{-3} \text{ wb/m}^2$. If the normal drawn to the plane of the coil makes an angle 30° with the field, then the torque on the coil is
 1) $3 \times 10^{-5} \text{ N-m}$ 2) $3 \times 10^{-3} \text{ N-m}$ 3) $3\sqrt{3} \times 10^{-5} \text{ N-m}$ 4) $3\sqrt{3} \times 10^{-3} \text{ N-m}$
- A moving coil galvanometer 'A' has 200 turns and a resistance of 100Ω . Another meter 'B' has 100 turns resistance 40Ω . All the other quantities are same in both the cases. The current sensitivity of A is
 1) double that of B 2) 2.5 times of B 3) 5 times of B 4) $\frac{1}{5}$ th of B

Numerical Value Type Questions

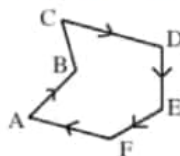
- A circular coil of 20 turns and radius 10cm is placed in a uniform magnetic field of 0.1T normal to the plane of the coil. If a current of 5A passes through the coil, then the torque on the coil is
- A coil in the shape of an equilateral triangle of side 0.02 m is suspended from a vertex such that it is hanging in a vertical plane between the pole pieces of a permanent magnet producing a horizontal field of $5 \times 10^{-2} \text{ T}$. Find the couple acting on the coil when a current of 0.1A is passed through it and the magnetic field parallel its plane ($\times 10^{-6} \text{ N-m}$)
- To increase the current sensitivity of a moving coil galvanometer by 50% its resistance is increased so that the new resistance becomes twice its initial resistance. By what percentage does the voltage sensitivity change ?

LEVEL-II (ADVANCED)

Straight Objective Type Questions

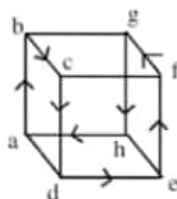
1. Find the magnitude of magnetic moment of the current carrying loop. Each side of loop is 10 cm long & current in the loop is $i = 2\text{A}$.

- a) 0.28 A-m^2
 b) 0.028 A-m^2
 c) 0.038 A-m^2
 d) 0.02 A-m^2



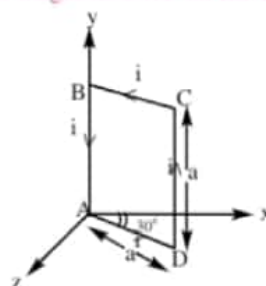
2. A conductor carries a constant current I along the closed path abcdefgha involving 8 of the 12 edges of length l . Find the magnetic dipole moment of the closed path

- a) $l^2 I \hat{j}$
 b) $l I \hat{j}$
 c) $\frac{l}{I} \hat{j}$
 d) $-l I \hat{j}$



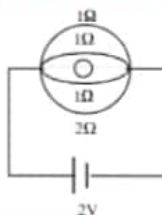
3. A square coil of side 'a' carrying current 'i' and is having one of its side AB parallel to y-axis and its plane is at angle $\theta = 30^\circ$ with x-axis (as shown). If a uniform magnetic field B exist in the region along $\hat{k}$ direction, then torque due to magnetic force on the coil is:

- a) $\frac{i a^2 B}{2} \hat{j}$
 b) $\frac{i a^2 B}{\sqrt{2}} (-\hat{k} + \hat{j})$
 c) $-\frac{i a^2 B}{2} \hat{j}$
 d) $\frac{i a^2 B}{2} \hat{i}$



4. Two circular conductors of radius 1 m each are arranged in mutual perpendicular planes with their diametric opposite points in contact as shown. The resistances of different semi circular parts are shown in figure. A small current carrying loop carrying current I_0 through it is arranged at the common center in stable equilibrium position and can be freely rotated about its diametric axis. The mass and radius of circular loop is m and r respectively. The time period of small oscillation of small loop is

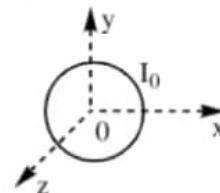
- a) $\sqrt{\frac{2\pi m}{\mu_0 I_0}}$
 b) $\sqrt{\frac{4\pi m}{\mu_0 I_0}}$
 c) $\sqrt{\frac{8\pi m}{\mu_0 I_0}}$
 d) $\sqrt{\frac{\pi m}{\mu_0 I_0}}$



More than One correct answer Type Questions

5. A uniform ring of mass m and radius R carrying current I_0 is lying in the x - y plane in vacuum with centre at origin. A uniform external magnetic field of strength $\vec{B} = B_0(2\hat{i} - 2\hat{j} + 5\hat{k})$ Tesla is switched on at $t = 0$ (here $B_0 = \text{constant}$). Gravity is neglected. If the ring can freely rotate, then

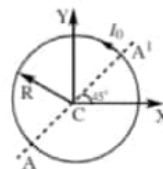
- a) Initial angular acceleration of ring is $\frac{4\sqrt{2}I_0B_0\pi}{m}$
 b) Initial angular acceleration of ring is $\frac{2\sqrt{2}I_0B_0\pi}{m}$
 c) The magnetic force acting on loop $2\sqrt{3}I_0\pi RB_0$
 d) The magnetic force acting on loop is zero

**Linked Comprehension Type Questions****Passage :**

A ring of mass m and radius r carrying current I_0 is lying in the x - y plane with centre at the origin. A uniform magnetic field of strength $\vec{B}_0(2\hat{i} - 3\hat{j} + 5\hat{k})$ T is applied in the region. If the ring can rotate about the axis AA' in x - y plane only.

6. Find initial angular acceleration

- a) $\frac{5\sqrt{2}I_0B_0\pi}{2m} \text{ rad/s}^2$ b) $\frac{2\sqrt{2}I_0B_0}{2\pi m} \text{ rad/s}^2$
 c) $\frac{\sqrt{2}I_0B_0}{3m} \text{ rad/s}^2$ d) $\frac{5\sqrt{2}I_0B_0\pi}{m} \text{ rad/s}^2$



7. Find the initial magnetic energy stored in the ring

- a) $-5\pi R^2 I_0 B_0 J$ b) $3\pi R I_0 B_0 J$ c) $-\pi R^2 I_0 B_0 J$ d) $2\pi R^2 I_0 B_0 J$

8. Find the force on the loop

- a) $-8\pi R^2 I_0 B_0 J$ b) $\frac{5\sqrt{2}I_0B_0\pi}{2m}$ c) zero d) $\sqrt{2}I_0B_0$

Matrix Matching Type Questions

9. In Column-I some current carrying elements are given along with the direction of magnetic field in the region. In Column-II information about the force and torque acting on elements are given. Match the entries of Column-I with the entries of Column-II

COLUMN - I

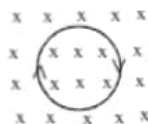
- A)
 B)
 C)
 D)

COLUMN - II

- p) Force = 0
 q) Force $\neq 0$
 r) Torque about O is zero
 s) Torque about O is non zero

Integer Type Questions

10. A solid non-conducting sphere of radius R and charge q (uniformly distributed) rotates about its diametric axis with constant angular velocity ω . The magnetic moment of the sphere is $\frac{qR^2\omega}{n}$. Find n
11. A circular current loop is placed in an uniform magnetic field of 1 tesla as shown. If the radius of loop is 1 m, the magnetic force on the loop is



KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

1) 2 2) 1 3) 3 4) 4 5) 2 6) 5cm 7) 0 8) 0.4

LEVEL-II

1) a 2) a 3) c 4) c 5) a 6) c 7) a 8) b
 9) b,d 10) bc 11) b,c,d 12) a 13) b 14) d 15) d 16) b
 17) c 18) A-p,q,r; B-p,q,r,s; C-r; D-p,q,s 19) 1 20) 4

EXERCISE-II

LEVEL-I

1) 1 2) 3 3) 2 4) 2 5) 2 6) 2 7) 2 8) 1
 9) 0.8 10) 420 11) 3

LEVEL-II

1) b 2) b 3) b 4) b,c 5) a 6) d 7) b,c 8) 2
 9) 5 10) 2 11) 1

EXERCISE-III

LEVEL-I

1) 4 2) 1 3) 1 4) 1 5) 4 6) 3.76 7) 12 8) 1.05

LEVEL-II

1) d 2) d 3) a 4) d 5) a 6) c 7) c 8) a
 9) a 10) a 11) 3

EXERCISE-IV

LEVEL-I

1) 2 2) 4 3) 3 4) 1 5) 14 6) 1.582 7) 0.327

LEVEL-II

1) b 2) a 3) d 4) c 5) b 6) c 7) d 8) b
 9) a,d 10) b,d 11) a,b,c,d 12) b,c,d 13) a 14) b
 15) c 16) 4 17) 3

EXERCISE-V

LEVEL-I

1) 3 2) 3 3) 1 4) 2 5) 1 6) 2 7) 4

LEVEL-II

1) d 2) c 3) abd 4) a,d 5) c 6) a 7) a 8) a
 9) b 10) c 11) A-p,q; B-p,r,s; C-p,s; D-p 12) 9 13) 8

EXERCISE-VI

LEVEL-I

1) 2 2) 1 3) 3 4) 2 5) 3 6) 7.2 7) 4 8) 9

LEVEL-II

1) b 2) b 3) d 4) a 5) b

EXERCISE-VII

LEVEL-I

1) 2 2) 1 3) 0 4) 0.866 5) 25

LEVEL-II

1) b 2) b 3) a 4) c 5) ad 6) d 7) a 8) c

9) A-q,r; B-p,r; C-p,s; D-p,s 10) 5 11) 0